

ANKIT KUMAR

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Northeastern University, Boston, MA

PROFESSIONAL SUMMARY

Applied Scientist and PhD candidate with demonstrated expertise in formal methods, security analysis, and software verification. Discovered and reported CVE-2022-47547 affecting Ethereum 2.0 networks, resulting in protocol-level patches. Author of 7 peer-reviewed publications (32 citations, h-index: 4) with 2 Best Student Paper Awards. Holds 3 pending patents from Amazon internships. Recognized internationally through invited presentations at IPFS Camp Lisbon and multiple ACL2 Workshops, with peer review service for premier venues (FMCAD, ICFEM, NFM, OOPSLA).

COMPENSATION & MARKET RECOGNITION

- **Amazon Applied Scientist Position (2026):** Total compensation of \$415,863 USD annually (year one)
- Market analysis demonstrates this compensation places candidate in **top 5-10% of Applied Scientists nationally**, significantly exceeding industry standards for recent PhD graduates
- Compensation is 65-110% above typical Applied Scientist base compensation, reflecting specialized expertise in formal methods, security analysis, and agentic AI systems
- Represents recognition of extraordinary ability in software verification and protocol security at enterprise scale

PROFESSIONAL EXPERIENCE

Applied Scientist Intern • Amazon Web Services • Santa Clara, CA • May 19, 2025 - Aug 22, 2025

- Pioneered **Agentic AI methodology** to automatically generate correct Rust code specifications from Java implementations
- Leveraged advanced **dataflow analysis** to eliminate hallucinations and improve code correctness
- Secured **two patent submissions (pending)** for innovative code verification and analysis techniques; earned return offer recognition in a full-time role
- Technologies: Java, Rust, Claude, Cline, Amazon Q, Agentic AI

Member of Technical Staff, Intern • Rivos Inc. • Portland, OR • Jun 7, 2023 - Sep 1, 2023

- Developed formal verification framework for **RISC-V hypervisor page walk algorithm security properties**
- Accelerated proof verification by 10x using strategic lemma transfer from high-level models
- Technologies: ACL2, ACL2 Sedan, Jaspergold

Applied Scientist Intern • Amazon • Minneapolis, MN • May 31, 2022 - Sep 2, 2022

- Engineered **taint-analysis framework** for enterprise-scale tracking of sensitive data across AWS infrastructure
- Secured **one patent submission (pending):** "Data Flow Analysis of Cloud-Based Software Applications"; earned return offer recognition
- Technologies: Java, Grails, GitHub, Shell Scripting

Software Development Engineer II • Olacabs • Bengaluru, India • 2012-2014

- Architected inaugural Olacabs Android consumer application, capturing **80% of total cab bookings**
- Reduced driver app crashes by **80%** through systematic debugging and optimization
- Implemented robust cashless payment and digital wallet systems
- Technologies: Java, Android, Git, Ruby on Rails

Associate • Morgan Stanley Advantage Services • Mumbai, India • 2010-2012

- Transformed legacy derivatives trading platform for multi-regional deployment
- Expanded platform reach to Hong Kong and Singapore, **doubling transaction capacity**
- Technologies: .NET, Java, Spring, Perforce

Upcoming Role: Applied Scientist • Amazon Web Services • Santa Clara, CA • Starting Jan 2026

EDUCATION

PhD in Computer Science • Northeastern University • Boston, MA • 2018-Expected 2026

- Thesis: Refinement-based reasoning of P2P protocols is feasible and useful
- **Research Impact:** 7 peer-reviewed publications, 32 citations, h-index: 4, i10-index: 1

PhD Research Contributions

- **1. P2P Protocol Verification**
 - Applied formal methods and refinement theory to **Gossipsub/Floodsub protocols**
 - Discovered and reported [**MITRE CVE-2022-47547**], a critical scoring function vulnerability affecting Ethereum and Filecoin networks, leading to real-world fixes
 - Published multiple peer-reviewed papers and presented work at IPFS Camp Lisbon 2022 and ACL2 Workshops (2023, 2025)
 - Related publications: [1], [2], [3]
- **2. Education: Automated Student Feedback Systems**
 - Developed **theorem-prover-backed automated feedback systems** for grading student submissions
 - Integrated with Gradescope to provide formal verification of student solutions and proofs
 - Enabled scalable and precise assessment of student work in formal methods courses
 - Related publications: [4], [5], [6]
- **3. Verification tools and algorithms: String Solvers**
 - Designed and implemented sound and theoretically complete **string equation solvers** for verification and formal reasoning tasks
 - Applied these solvers to both academic research and practical software verification problems
 - Related publications: [7]

M.Tech in Computer Science • IIT Kanpur • Kanpur, India • 2015-2017

B.Tech in Electrical Engineering • IIT (BHU) Varanasi • Varanasi, India • 2006-2010

SECURITY RESEARCH & IMPACT

- **CVE-2022-47547 Discovery - Critical Infrastructure Security Impact:**
 - Identified and reported critical vulnerability in GossipSub 1.1 protocol, the backbone of Ethereum 2.0's peer-to-peer consensus mechanism

- **Systems Affected:** Ethereum 2.0 Beacon Chain (securing 1M+ validators and \$100B+ in staked assets as of 2024), Filecoin network, and IPFS infrastructure
- **Severity:** CVSS Score 5.3/10 (Medium) - Network-based attack vector with low complexity, affecting message propagation integrity
- **Official Recognition:** Published in National Vulnerability Database (NVD) on December 19, 2022; Last modified April 17, 2025
- **Industry Impact:** Coordinated disclosure with MITRE (CVE Program); resulted in protocol-level patches deployed across Ethereum ecosystem, affecting global blockchain infrastructure
- **Technical Contribution:** Vulnerability allows malicious peers to maintain positive reputation scores while continuously misbehaving, enabling sustained network manipulation attacks
- **Verification Method:** Discovered through formal mathematical modeling and proof-based analysis techniques, demonstrating practical value of formal verification in production systems
- Publication documenting discovery: IEEE Symposium on Security & Privacy 2024 (premier security venue)
- **Formal Verification Research Impact:** Applied mathematical proof techniques to analyze critical peer-to-peer protocols, revealing previously unknown attack vectors in production systems used by millions
- **Academic Contributions:** 7 peer-reviewed publications (32 citations, h-index: 4) demonstrating formal verification methodologies for distributed systems and blockchain protocols
- **Intellectual Property:** 3 patent applications pending with US Patent Office covering novel techniques in dataflow analysis, code verification, and security analysis
 - US Patent Application (Pending): "Data Flow Analysis of Cloud-Based Software Applications"
 - Two additional patent applications (Pending): Novel approaches to AI-driven code generation and verification

PROFESSIONAL SERVICE & COMMUNITY CONTRIBUTIONS _____

- **Peer Review Service:** Reviewed papers for FMCAD 2025, ICFEM 2023, and NFM 2024 (premier formal methods conferences)
- **Artifacts Evaluation:** Served on OOPSLA 2024 and ICFP 2023 Artifacts Evaluation Committees (top-tier programming languages conferences)
- **Conference Leadership:** Invited to chair SIGCSE TS Conference, Canada, 2023 (leading computer science education conference)
- **Volunteer Service:** Student volunteer at FLoC 2022, Haifa, Israel (federated logic conferences)

PUBLICATIONS _____

1. **A Formalization of the Correctness of the Floodsub Protocol**
ACL2 Workshop 2025, Austin, TX • Best Student Paper Award • [\[PDF\]](#)
2. **Formal Model-Driven Analysis of Resilience of GossipSub to Attacks from Misbehaving Peers**
IEEE Symposium on Security and Privacy 2024, San Francisco, CA • [\[PDF\]](#)
3. **Verification of GossipSub in ACL2s**
ACL2 Workshop 2023, Austin, TX • Best Student Paper Award • [\[PDF\]](#)

4. **Automated Grading of Automata with ACL2s**
Electronic Proceedings in Theoretical Computer Science 2023 • [\[PDF\]](#)
5. **Proving Calculational Proofs Correct**
ACL2 Workshop 2023, Austin, TX • [\[PDF\]](#)
6. **Calculational Proofs in ACL2s**
arXiv preprint 2023 • [\[PDF\]](#)
7. **Mathematical Programming Modulo Strings**
Formal Methods in Computer Aided Design 2021, Virtual Conference • [\[PDF\]](#)

INVITED TALKS & RESEARCH DISSEMINATION ---

- **IPFS Camp Lisbon 2022** - Invited International Presentation
"Formal Verification of Gossipsub and Network-Level Vulnerabilities"
Hosted by Protocol Labs (creators of IPFS/Filecoin), Lisbon, Portugal
Archived on Protocol Labs official YouTube channel: [\[Video Link\]](#)
Reached international audience through organizer's platform
- **Syracuse University 2024** - Invited Research Talk
Presented dissertation research on formal methods and P2P protocol verification
- **Boston Computation Club** - Invited Technical Presentation
"Mathematical Programming Modulo Strings"
Archived on Boston Computation Club official channel: [\[Video Link\]](#)
Technical community presentation demonstrating research accessibility and impact

TECHNICAL SKILLS ---

- **Programming Languages:** C, C++, Java, Rust, Lisp, ACL2, ACL2 Sedan, Coq, Haskell, Ruby
- **Development Tools:** Git, GitHub, Emacs, LaTeX, Shell Scripting, Perforce
- **Frameworks & Technologies:** Spring, .NET, Grails, Ruby on Rails, Android
- **AI & Machine Learning:** Claude, Cline, Amazon Q, Agentic AI
- **Verification Tools:** Jaspergold, ACL2, ACL2 Sedan
- **Specializations:** Formal Methods, Theorem Proving, AI Agents, Software Verification, Security Analysis, Dataflow Analysis, Taint Analysis

AWARDS & HONORS ---

- **Best Student Paper Award, ACL2 Workshop 2025**
Consecutive Best Student Paper recognition demonstrating sustained research excellence (First researcher to receive this honor multiple times)
- **Dissertation Completion Fellowship, Summer 2024**
Competitive university-wide fellowship recognizing outstanding dissertation research
- **Best Student Paper Award, ACL2 Workshop 2023**
Selected from competitive field of submissions; ACL2 is premier venue for automated reasoning research
- **Student Travel Fellowship, PLMW at POPL 2019**
Competitive fellowship for Programming Languages Mentoring Workshop at top-tier conference
- **First Prize, Droidcon India Hackathon 2013**
National-level Android development competition